

**LEADERSHIP IN THE AGE OF ARTIFICIAL INTELLIGENCE: NAVIGATING
THE MORAL AND ETHICAL COMPLEXITIES OF AUTONOMOUS SYSTEMS IN
THE DEFENCE FORCES.**

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STUDENT DECLARATION

1. I certify that this thesis does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma at any university and that, to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.
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ABSTRACT

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This thesis analyses the ethical challenges and leadership implications posed by the integration of Artificial Intelligence and autonomous systems into the Defence Forces' command and control structures. Recognising the strategic backdrop of the evolving geopolitical landscape and dynamic nature of cyberspace, this research examines how military leadership can adapt to maintain ethical values, accountability and operational effectiveness as it seeks to embrace emerging and disruptive technologies. Through detailed analysis, case study evaluations and critical engagement with the latest research, particularly targeting and cyberspace operations, the study reveals a persistent tension between technical efficiency and ethical responsibility. This research identifies three central challenges: the erosion of human judgement in military decision-making, the attribution problem regarding cyberspace operations and autonomous systems, and lacunae surrounding leadership doctrine in addressing these challenges. Case studies illustrate the consequences of reduced human agency in targeting decisions. The thesis presents the Collateral Effects Estimate Process as a viable tool to assist the commander in assessing the impact of AI-enabled operations. Key recommendations include the creation of a dedicated Artificial Intelligence oversight section within the Joint Cyber Defence Command, a revision of the leadership framework to include human-AI interaction, and the formulation of a policy for accountability that will be consistent with International Humanitarian Law. Artificial Intelligence has the potential to be either a strategic multiplier or a liability; this thesis argues for a proactive response to ensure that operational benefits are consistent with ethical values.

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GLOSSARY OF ACRONYMS AND ABBREVIATIONS

ACT	NATO Allied Command Transformation
AEGIS	American Integrated Naval Weapons System
AI	Artificial Intelligence
AWS	Autonomous Weapon Systems
CCDCOE	NATO Cooperative Cyber Defence Centre of Excellence
CEEP	Collateral Effects Estimate Process
CSIRT	Computer Security Incident Response Team
CNI	Critical National Infrastructure
CO	Cyberspace Operations
COIN	Counter Insurgency Operations
COTS	Commercial-off-the-Shelf
CyCon	International Conference on Cyber Conflict
C2	Command and Control
DARPA	Defence Advanced Research Projects Agency
DDOS	Distributed Denial of Service
DF	Defence Forces (Ireland)
EDA	European Defence Agency
EDT	Emerging and Disruptive Technologies
EU	European Union
GNSS	Global Navigation Satellite System
GRU	Russia's Main Intelligence Directorate
HSE	Health Service Executive
IDF	Israel Defense Forces
IHL	International Humanitarian Law
IMSI	International Mobile Subscriber Identity
IO	Information Operations
ISR	Intelligence, Surveillance and Reconnaissance
JCDC	Joint Cyber Defence Command
LAWS	Lethal Autonomous Weapon Systems
ML	Machine Learning
MDO	Multi-Domain Operations
NATO	North Atlantic Treaty Organisation
NCSC	National Cyber Security Centre
NSE	Non-Strike Entities
OODA	Observe, Orientate, Decide, and Act
PMESII	Political, Military, Economic, Social, Infrastructure and Information
SA	Situational Awareness
SCADA	Supervisory Control and Data Acquisition
SCEPVA	Sovereign Cyber Effects Provided Voluntarily by Allies
TTP	Tactics, Techniques and Procedures
VUCA	Volatility, Uncertainty, Complexity and Ambiguity
WLAN	Wireless Local Area Network
5G	Fifth-Generation Technology for Mobile and Internet Services

INTRODUCTION

“Quemadmodum gladius neminem occidit, occidentis telum est”

Introduction

The integration of artificial intelligence (AI) into military operations indicates a significant evolution in modern military conflicts, raising profound operational questions and ethical concerns. As militaries adopt emerging and disruptive technologies (EDT), the Defence Forces (DF) will face significant challenges, particularly regarding Command and Control (C2) systems, autonomous targeting, and managing a Collateral Effects Estimate Process (CEEP). These areas involve serious decisions where the delegation of authority to AI impacts mission assurance and may have a significant cost in human lives. AI denotes a basket of technologies whose common attribute is the capability to simulate human cognition, particularly the human brain’s ability to adaptively reason, learn and autonomously undertake appropriate actions in response to a given environment (Whyte, 2020, p. 219). The technologies of AI might be thought of as existing across three main categories: sensing and perception, movement, and machine learning (ML). By far the most significant area within this category is that of ML, the scientific study and development of approaches to pattern recognition and knowledge construction without pre-programmed instructions on how to interpret data. (p. 220).

This thesis seeks to determine how military leadership can address these challenges and maintain operational effectiveness. It analyses the evolution of leadership frameworks and promotes accountability, transparency, and human oversight, focusing on C2, targeting procedures, and a CEEP. This study offers practical recommendations for the DF as it navigates technological advancements and ethical responsibilities. By contributing to discussions on leadership and technological shifts, it offers a framework to guide the DF strategic adaptation of AI.

Literature Review

The increasing use of AI in military operations has garnered significant academic and policy interest, and the existing literature has explored the technological aspects of AI systems, their operational applications, and the strategic advantages they offer (Mayer, 2023) (Yan, 2020). Studies on C2 systems highlight AI’s potential to enhance decision-making, particularly in high-stress scenarios, by providing rapid data analysis and predictive modelling capabilities (Davis, 2022). Nonetheless, critical gaps remain in addressing the ethical and moral challenges

that arise from their deployment. Trust collaboration between human operators and AI systems is essential for effective decision-making but remains underexplored. Additionally, an over-reliance on automated systems poses significant challenges, including their inability to adapt to nuanced situations and real-time ethical considerations (Mayer, 2023).

Furthermore, research on autonomous targeting has primarily focused on technical precision and operational efficiency and limits its attention to the moral implications of delegating life-and-death decisions to algorithms (Yan, 2020). Research often fails to address the implications of removing human agency from such decisions and focuses instead on optimising system performance. This narrow focus neglects the broader ethical discourse surrounding accountability and the value of human oversight in targeting processes. While ethical frameworks from the military regarding AI exist, such as those proposed by the North Atlantic Treaty Organisation (NATO) and the European Defence Agency (EDA), they remain underdeveloped and lack practical applicability in a dynamic operational context (Ertan, 2022). These frameworks often fail to address the unintended consequences of AI deployment, including biases embedded in algorithms and the challenges of ensuring accountability in complex environments. Literature addressing collateral effects often omits discussions about unintended biases and the challenges of ensuring accountability in complex environments (Davis, 2022).

This research highlights the need for a comprehensive approach that integrates ethical considerations with practical leadership strategies. The DF requires frameworks that address operational effectiveness and uphold moral values and ethical principles. Through combining insights from existing research with practical examples, this study aims to contribute to the development of leadership strategies that enable the DF to cope with both technological advancement and ethical responsibility.

Research Lacunae

The integration of AI into military leadership and operations is the subject of increasing academic and professional scrutiny; however, significant gaps remain in this field of research. While considerable attention has been paid to the technical capabilities and strategic advantages of AI systems, there is a marked absence of studies focusing on the ethical and moral challenges faced by military leadership when engaging with EDTs. This thesis seeks to address these gaps by examining specific dilemmas arising from the use of AI in C2, decision-making, targeting and a CEEP within the DF.

One notable gap is the limited exploration of how autonomous decision-making systems impact the principle of accountability in military operations. The delegation of critical decisions to AI systems raises questions about who bears responsibility for military actions taken and effects delivered by autonomous platforms, particularly in scenarios where civilians could be harmed or missions fail. Some studies address the operational benefits of AI; however, only a few have critically assessed the ethical frameworks necessary for practical guidance, leaving a significant vacuum for military leaders. Another gap lies in the study of human-AI interactions within military contexts. Existing literature often neglects the complexities of trust collaboration between human operators and AI systems, which is essential for effective decision-making. This oversight is particularly significant given the increasing reliance of the DF on data-driven technologies in dynamic and high-pressure environments – not least in its efforts to achieve Level of Ambition 2 as per the Detailed Implementation Plan (DOD, 2023).

Furthermore, there is a scarcity of research examining the long-term implications of AI integration for military culture and leadership development. The DF must adapt to the immediate challenges posed by AI but also consider how these technologies will reshape leadership paradigms and ethical considerations in the future. In addressing this research lacuna, this thesis aims to provide a comprehensive analysis of the ethical and operational challenges of AI and military leadership. The insights generated will contribute to both academic and practical policy development, offering actionable recommendations for DF leadership as it navigates the evolving landscape of AI-enabled operations.

Research Questions

This thesis is guided by three core research questions designed to address the identified gaps and explore the ethical and operational dimensions of AI integration in military leadership:

- 1. What are the key ethical challenges that AI and autonomous systems pose for military leadership within the DF?*

This question investigates the moral and ethical dilemmas associated with the deployment of AI-enabled systems, particularly in C2 systems, targeting procedures and CEEP.

- 2. How can military leadership frameworks adapt to ensure accountability, transparency, and human oversight with the integration of AI systems?*

This question focuses on the practical strategies and leadership theories necessary to navigate the complexities of human-AI interactions and decision-making.

3. *What leadership strategies can be developed to maximise operational benefits of AI while mitigating these ethical risks?*

This question examines the balance between leveraging AI's technological advantages and maintaining ethical standards in military operations.

By addressing these questions, the research aims to provide actionable recommendations for the DF, ensuring that military leaders are equipped to meet the dual demands of technological innovation and ethical responsibility.

Sources

This thesis draws on a diverse range of sources to provide a comprehensive understanding of the ethical and operational challenges associated with integrating AI into military leadership. The primary sources include official DF doctrines and leadership frameworks, which provide insights into current principles and practices (Defence Forces, 2023). These foundational documents are crucial for evaluating how current leadership paradigms can adapt and incorporate AI technologies into the DF.

Secondary sources consist of peer-reviewed academic journals, books, reports, and industry publications exploring AI's role in military operations. Notable contributions include studies on human-autonomy teaming, trust collaboration and ethical frameworks for autonomous systems (Ertan, 2022) (Mayer, 2023). These sources contextualise the challenges faced by the DF within broader global trends.

Supplementing this research are case studies from contemporary conflicts, including the use of AI-enabled systems in the Russia-Ukraine war and the conflict in Gaza. These studies provide concrete military examples that support the theoretical discussion (Bergengruen, 2024). Furthermore, international benchmarks, including guidelines from NATO and the EDA, compare the DFs' efforts against best international practice (NATO, 2024). Combined, these sources support a robust and multidimensional analysis, ensuring the research is both theoretically grounded and practically relevant.

Methodology

The methodical approach to this thesis is rooted in qualitative research, leveraging an arts and humanities framework to critically analyse the moral and ethical challenges of AI and military leadership. This approach enables comprehensive exploration of the intersection between leadership theories and EDTs, addressing complex issues in depth. To achieve this, the research applies a combination of the following methods:

1. **Systematic Literature Review (SLR):** systematic analysis of peer-reviewed academic, governmental, and industry sources to establish a theoretical foundation. This includes examining key leadership doctrines and ethical frameworks related to AI integration and military operations. The SLR framework ensures transparency, traceability, and reduced bias (Tranfield, et al., 2003).
2. **Case Studies:** In-depth analysis of real-world applications of AI in military contexts, such as its use in C2 systems during recent conflicts. These studies provide practical insights into the ethical and operational implications of AI technologies.
3. **Comparative Analysis:** benchmarking DF strategies and practices against international standards and best practices from organisations such as NATO and the EDA. This approach highlights gaps and opportunities for improvement.
4. **Thematic analysis:** identifying recurring themes and patterns across the collected data to draw meaningful conclusions about the ethical and leadership challenges posed by AI.

Through integrating theoretical and practical perspectives, this methodology facilitates a deeper appreciation for how military leaders can adapt to the evolving landscape of AI-driven operations while upholding ethical standards.

Thesis Outline

Following the methodology above, this thesis is structured into three main chapters which explore the solutions to the research questions.

Chapter One: AI and Command and Control Systems and Decision-Making Processes explores the integration of AI in decision-making frameworks, focusing on ethical dilemmas and operational challenges.

Chapter Two: AI and Military Targeting analyses the use of AI in targeting systems, highlighting moral implications, accountability concerns and leadership's role in ensuring ethical practices.

Chapter Three: AI and Collateral Effects Estimate — Accountability and Attribution examines the unintended consequences of AI deployment, including algorithm biases, civilian harm, and the long-term impact on military culture.

Each chapter integrates case studies, literature, and comparative insights to provide actionable recommendations for the DF.

CHAPTER ONE: AI AND COMMAND AND CONTROL SYSTEMS AND DECISION-MAKING PROCESSES

“Who’s in Charge?”

1.1 Introduction

The main themes of this chapter are an overview of AI in military C2 systems – why militaries need to integrate AI into command structures, the significance of AI in modern warfare (exploring its role in decision-making), and ethical dilemmas – speed versus accuracy, autonomy versus oversight, and ethical concerns. The recent development and meteoric rise of AI, along with its vital importance to autonomous systems, have presented states and militaries with both opportunities and threats to their national security (Khanal, et al., 2025). Along with an increasingly complex geopolitical landscape and (in)security environment, the integration of AI into military operations raises serious moral and ethical considerations for leaders within the military. A 25-year background in communications and information systems and, more recently, in cyberspace has aroused the author’s interest in the role and use of AI in the military. The original designers and advocates of the development of AI technology, focused as they were on the ‘classical’ or ‘statistical’ methods, certainly had little or no vision that AI would someday be used in armed conflict (Boden, 2018, p. 15). This viewpoint may have been naïve, however, as DARPA has underwritten several AI-related projects (Fouse, et al., 2020).¹ There is some nascent concern regarding the uses and dilemmas that AI technology may present – particularly in the process of targeting by Autonomous Weapon Systems (AWS).

1.2 Foundations of Command and Control

Before addressing the ethical concerns regarding AI and AWS, it would be prudent to establish our understanding of military command and its often-espoused quality of leadership. Additionally, it will be beneficial to distinguish between command, leadership and management. Military command is nested in regulation and grants military authorities the legal basis for the legitimate use of force (Defence Forces, 2023, pp. 3-1). The basis of leadership, as the DF champions it, is based on the core organisational values of physical courage, moral courage, loyalty, selflessness, respect, and integrity (pp. 2-1). It is worth noting that although moral courage is defined, explained and illustrated with a vignette in the leadership doctrine, there is a surprising lack of discussion regarding ethics (2023, pp. 4-3). Does this suggest that

¹ Defence Advanced Research Projects Agency <https://www.darpa.mil/>

the ethical considerations of the military are entirely dependent on the moral standing of its individuals?

Shamir suggests while ‘mission command’ is best suited for unconventional warfare scenarios, for example COIN, the concept remains valid for digital C2 systems which “ostensibly increase micromanagement” (2011, p. 4).² While many argue that management is an important element of leadership, with those that do not being considered distant and disconnected, it is important to stress leadership is an essential part of military command (Mintzberg, 2009). Indeed, DF doctrine describes leadership as the ability to provide “purpose, direction and motivation” to people, emphasising that in the military environment it involves “moral and ethical complexities rarely experienced in civilian life” (Defence Forces, 2023, pp. 3-1). The Comprehensive Operational Planning Directive, the primary planning tool for NATO and partner militaries, states that “clear command relationships are central to effective operations and organisations” (SHAPE, 2021, pp. 4-34). Decision-making implies that leaders have access to information and the means with which to relay their orders, hence the importance of C2 systems, particularly in the modern engagement space. A key consideration for future leaders and current decision-making processes concerns the successful, or otherwise, integration of AI. While traditional military leadership and, by implication, C2 systems are based on organisational values, AI’s reliance on algorithms and data-driven appetites may create a disconnect between the past and the present (Defence Forces, 2023, pp. 4-2). Trust is a fundamental element of military leadership and stewardship. Mayer (2023) notes that training with autonomous systems has the potential to encourage trust in human teaming.

The DF will have to grapple with the following four rationales when it comes to the use of AI systems in the military: ethics, legality, stability, and safety (Maas, 2019). The DF has articulated an understanding to embrace the “effective adaptation to new technologies” (Defence Forces, 2023, pp. 8-7). Mayers argues that the proliferation of AI in military operations will lead to more interaction between humans and intelligent machines (2023, p. 522). A key aspect of leadership is the essential element of decision-making. However, military officers are often accused of “analysis paralysis” (Gustafson & Davies, 2019, p. 26). A question we need to consider throughout this thesis is: are the DF values a sufficient guide for the ethical use of AI in the military?

² Counter Insurgency Operations

AI will impact every aspect of modern society, and the military will be no exception to this evolution. AI's role in military strategy is increasingly recognised through international frameworks such as the European Union (EU) (2024) and NATO's (2024) policies on the use of AI in the military. In mid-2024, the EU published its Artificial Intelligence Act (EU, 2024). A critical component of this legislation is the pursuit of explainable AI, particularly for its use in high-risk systems (p. 59). Although there is no specific mention of AI or EDTs in NATO's Strategic Compass (2022), EDTs are recognised by NATO ACT as being critical enablers of its five warfighting capstone concepts (NATO ACT, 2021).^{3 4}

In Ireland, and of particular importance to the DF, are the recent guidelines by the National Cyber Security Centre (NCSC) and Department of Public Expenditure and Reform (DPER). The latter states that the government has committed to promoting the ethical use of AI (DPER, 2024, p. 4), and the former, along with a security caution, has issued a number of practical recommendations (NCSC, 2023, p. 5). Each of these documents will guide the DF. The DF Leadership Doctrine states that the “contemporary operating environment is recognised...as being in a constant state of flux, resulting in new missions, new technologies...and new concepts continually coming on stream” (2023). This paper emphasises a clear appetite to anticipate the changes to come about as a result of the development of EDTs and the presumed integration of AI into DF C2 systems. However, it is worth noting that AI and quantum computing only get a cursory mention in the recently updated DF Cyber Defence Strategy 2024 (Defence Forces , 2024).

Ertan & Gray postulate that AI's impact on strategy can be framed through the lens of revolutionary military affairs asking; is AI really a disruptive technology, or is it just being normalised now (2021, p. 13)? Strategy can often be distilled down to a simple equation composed of Ends, Ways and Means, but where will AI be best positioned to influence strategy – should this new technology be viewed as a service or as a resource (Ministry of Defence, 2014, p. 8)? Strategy can also be understood via its utility – the value of a resource or action, less its costs, determines its utility in strategy (Knorr, 1996). Clausewitz's concepts of “friction” and the “fog of war” emphasise the unpredictable nature of conflict, and it is important that military commanders learn how to navigate uncertainty and the role of chance in the modern engagement space. Hoffman suggests that friction and fog will remain, and the

³ Emerging and Disruptive Technologies

⁴ Allied Command Transformation

introduction of autonomy will bring new forms of friction — despite advances in technology and computing power, battlespace uncertainty will persist (2018, p. 21).

Though it seems that the age of AI has arrived abruptly, it is important to recognise that it has been developing for over half a century, since the cracking of the codes of the Enigma machine.⁵ Early developments of AI were often fixated on the possibility of it being able to mimic, and then become indistinguishable from, human actions and decisions, with Alan Turing (1950) developing the “Turing Test” modelled on a Victorian parlour game which became the *de facto* standard (Wilks, 2023, p. 18). It was only much later that developers and philosophers began to articulate (and struggle with) the ethical dimensions of autonomous systems making decisions. For example, the “runaway train scenario” remains unresolved (Wilks, 2023, p. 192). Zhou *et al.* outline that AI-based decision-making systems and military operations are becoming increasingly complex (2024). They note that AI-based decision-making systems are now employed at various command levels, from strategic planning to tactical targeting. While these systems offer enhanced efficiency, they also affect traditional military judgement and decision-making. Such an effect necessitates a reassessment of C2 structures to incorporate human oversight effectively.

The asymmetry of current conflicts highlights how challenging military operations have become and the increased security dilemmas facing the military commander. The ability for a military to remain flexible and adaptable to changing and unforeseen circumstances (the fog and friction) illustrates the value of innovation (Defence Forces, 2023, pp. 5-4). This implies that it is imperative that militaries accept AI into C2 systems and decision-making processes, including weapon systems, to afford the commander the insights needed to successfully achieve their objective(s). AI, if properly used, can serve as an effective tool to reduce the friction and alleviate the fog of modern conflict. However, the use of AI by an adversary is only going to increase the VUCA a commander faces in the modern engagement space — the adversary “always gets a vote”.⁶

The PMESII framework is an apparatus for military staff and planners at the strategic, operational and component command levels to assist their understanding of the modern engagement space (SHAPE, 2021, pp. 1-13).⁷ The successful integration of AI into this

⁵ A cipher device used by the German armed forces during the Second World War to encrypt and decrypt messages, and first decrypted by the Polish in 1932.

⁶ Volatility, uncertainty, complexity, and ambiguity

⁷ Political, Military, Economic, Social, Infrastructure and Information domains as recognised by NATO

framework will have enormous benefits for military and operational planners. However, there will be challenges in showing that AI-generated recommendations align with strategic objectives and directions, and do not inadvertently cause increased tensions in the conflict or crisis. Nominally absent from PMESII but integral to a commander's understanding of a crisis are the legal and technological dimensions. Greipl (2024) is critical of the binary conversation regarding AI systems and military decision-making, claiming that this perspective oversimplifies the complexity of their integration into conventional C2 structures. There is a view that AI systems are inherently better or worse than humans under International Humanitarian Law (IHL) conditions, but this neglects the nuanced reality where humans and machines collaborate throughout the technological readiness cycle.

Nearly a decade after being recognised as a domain of operations, and despite its critical role in day-to-day business and deployments, many militaries still struggle to successfully integrate cyberspace operations (COs) into their planning processes (CCDCOE, 2020). Tellingly, among a plethora of courses available at the NATO CCDCOE, one of the only three NATO ACT-approved courses delivered there is 'Integrating Cyber Considerations into Operational Planning'.^{8 9} It is challenging to integrate COs into operational planning, along with the education, training and resources involved in developing cyber "assets". Among the main challenges at the operational level is that nations keep cyberspace capabilities (and approvals for their use) at the strategic level. Outside of threat-hunting operations confined to its own networks, NATO is entirely reliant on the SCEPVA mechanism for the delivery of COs and their operational effects (CCDCOE, 2020, p. 37).¹⁰ Therefore, it may be suggested that militaries might view AI in a similar manner and reserve this technology as a national asset — a strategic resource analogous to nuclear?

Popa (2022) stresses that AI systems, while capable of enhancing tactical efficiency, raise significant ethical concerns, particularly in the deployment of lethal autonomous weapon systems (LAWS) (p. 392). The potential for a fully autonomous system to make life-or-death decisions without human intervention necessitates strict regulatory oversight. For instance, the Israel Defense Forces (IDF) deployment of an AI-coordinated drone swarm in Gaza demonstrates both the capability and ethical challenges of such technologies (p. 387). The

⁸ Cooperative Cyber Defence Centre of Excellence <https://ccdcoe.org/>

⁹ <https://ccdcoe.org/training/integrating-cyberspace-considerations-into-operational-planning-september-2025/>

¹⁰ Sovereign Cyber Effects Provided Voluntarily by Allies

question of whether such technologies should make decisions without human oversight remains deeply contentious and emphasises the importance of stricter international regulations.

Abraham (2025) addresses the IDF's use of AI systems such as Lavender, which has drastically reduced the demand for human monitoring as officials spend less time confirming targets. Given the alleged ten percent mistake rate that has resulted in the bombing of non-combatants, this dependence on AI-generated targeting raises ethical concerns. These tactics draw attention to the natural dangers connected to using AWS and AI technology in military combat operations. Davies's article reveals how the IDF uses an AI system, the Gospel, which has transformed targeting decisions into a high-speed autonomous process (2023). However, this system often prioritises efficiency over precision. AI recommendations have been overused and there has been minimal time verifying targets. Such a juxtaposition raises concerns about biases and ethical accountability in military decision-making processes.

The use of AI in military C2 systems raises concerns about accountability, human oversight and real-time decision-making. Michael Schmitt, a contributor to the Tallinn Manual, highlights the significant concept of humans being '*on the loop*', '*out of the loop*' or '*in the loop*'. This illustrates the critical importance of understanding the role of human agency in the deployment of AWS (NATO CCDCOE, 2021, p. 129). Ethical issues abound, especially in targeting and the use of AWS, and these will be covered in more detail in Chapter Two.

Other important considerations are transparency and explainability. The EU's AI Act seeks to develop explainable AI (EU, 2024, p. 19). However, should this really be a priority considering the limitations humans possess regarding their understanding of their personal agency and decision-making? The field of behavioural economics is full of studies detailing the cognitive biases and heuristics, flawed or otherwise, that humans use to rationalise their decision-making (Kahneman, 2010). Of course, one of the major considerations for the use of AI is who will programme the algorithm? This is an important question, and a number of other questions immediately fall from it: will the DF simply procure a COTS version of AI, or are they going to develop their own AI system?¹¹ If it is the latter, then who in the DF will programme it, and what values will be guiding them – their own sense of moral courage or DF ethical considerations (Defence Forces, 2023)?

¹¹ Commercial-off-the-Shelf

1.3 Operational and Tactical Applications of AI in C2 Systems

If an effective strategy is about measuring the utility of resources (Knorr, 1996) and effective leadership is about coping with changing circumstances (Northouse, 2016, p. 319), then the modern-day military commander should welcome AI as a tool that will help them to navigate and act decisively in an engagement space heavily affected by VUCA conditions. The most significant advantages of using AI during the planning of operations will be its capacity to process information quickly, therefore accelerating decision-making cycles, and this may take the form of multiple “narrow” AI tools, each focused on a particular task (Davis, 2022). Ultimately, in critical environments where rapid response times and quick decisions need to be met, AI-enabled systems must support commanders by analysing complex scenarios and identifying, and recommending, optimal courses of action within the operational planning process, at least in theory (Ertan & Gray, 2021, p. 14).

As mentioned, the concepts of bounded rationality and cognitive biases severely influence human decision-making (Kahneman, 2010). A clear example of this is the USS *Vincennes* naval ship involved in crossing into the territorial waters of Iran, and resulted in the downing of Iran Air Flight 655 (Fisher & Kingma, 2001, p. 113). How could AI improve human decision-making? Would the use of the currently deployed AEGIS Combat System on board eighty-four US Navy ships (US Navy, 2021) make for improved decision-making?¹² The OODA loop is a valuable tool for operators involved in real-time decision-making (Bryant, 2006).¹³ It will be vital for the utility of AI to be integrated into this loop for it to be of value in military operations, particularly at the tactical level and for those engaged in kinetic activity. As Schmidt advises, any involvement of an autonomous system must have a human being either in or on the decision-making loop (NATO CCDCOE, 2021, p. 129).

The military must act decisively, particularly during kinetic engagements — that is why so much time and effort is spent on planning, preparing, training and professional military education (O’Hara, 2016, p. 20). Can AI be effective in a VUCA environment? How fallible would it be? The ability of AI to conduct predictive analysis has the potential for advancing decision-making processes, particularly within the OODA loop. Pattern recognition in adversary behaviour, which involves predicting possible threats, implies that AI can significantly improve military situational awareness (SA) and offer appropriate response

¹² American integrated naval weapons system

¹³ Observe, Orientate, Decide, and Act

options. However, it is imperative to address concerns regarding data reliability and biases within the algorithm to avoid reaching poor, ineffective conclusions. There are many examples of generative AI models being liable to ‘hallucinate’ when faced with obscure tasks or ambiguous data which may overload their operating capacity (IBM, 2023).

Kajander *et al.*’s (2020) article from CyCon discusses the implications of Common Article 1 of the Geneva Conventions on states supplying and using AWS (p. 79).¹⁴ Common Article 1 requires states to respect and ensure respect for the Convention in all circumstances, including a positive application for third countries to prevent violations. The article examines the positive obligations and impact on states applying AWS in a conflict. These obligations include ensuring that the AWS cannot violate Geneva Conventions at the manufacturing stage. This implies that if the DF were to procure an AWS, it would need to have advanced procurement and capability development processes to ensure that the original equipment manufacturer is writing code that will be compliant with the Geneva Convention. However, as the article points out, because AWS systems are somewhat uncontrollable and unpredictable – as they will have to have the ability to learn; rather than only carrying out pre-programmed commands – the supplying state may be obligated to maintain a permanent tether to the supplied AWS (p. 83). Tethering the supplied AWS may extend beyond ensuring compliance with IHL and may multiply the leverage of the supplying state by turning the system into a “cyber mercenary” (p. 91).

This case is a clear example of the dilemmas which will face military leaders and commanders as they seek to procure and employ AI and AWS for military purposes. How can the military ensure compliance with regulations when there is so much uncertainty in an AI system? Kajander *et al.* propose that a tethered AWS would have a remote ‘cyberlink’ (p. 92) to the supplying state, which may be activated anytime, thus involving them in the conflict. They suggest this issue of tethering could be broken down on a case-by-case basis, where a relatively low risk of causing violations would not be under a regulatory requirement but a high-risk killer drone would be (p. 94). Nonetheless, all these considerations are underpinned by the cybersecurity of the AI system, the supplying manufacturer and the military itself (Ertan & Gray, 2021).

Ertan outlines several extensive cybersecurity issues regarding the deployment of AI, such as the susceptibility to data poisoning, where malicious actors can manipulate training

¹⁴ Annual International Conference on Cyber Conflict held in Tallinn, Estonia: <https://ccdcoe.org/cycon/>

data to cause an AI misclassification (2022, p. 51). Given the vulnerability of Ireland to cyberattacks following the HSE incident (PwC, 2021) and the inertia surrounding developments aimed at improving national response to same (Staunton, 2011), it is important to highlight these known cyber vulnerabilities with AI systems. The same security of AI systems depends on data integrity, algorithm transparency, and risk mitigation strategies. Popa draws attention to a major concern with military AI and its vulnerabilities to cyberattacks (2022, p. 390). They explain how adversaries can exploit these systems through tactics such as feeding false data (“easily fooled”) or causing them to misclassify critical information (p. 390). This kind of manipulation could have devastating consequences, making cybersecurity a key priority in military AI deployment as these technologies become more widespread. Ertan (2022) explains that many AI applications in most contexts lack standardised security frameworks, leaving them vulnerable to cyber threats. Therefore, securing AI-driven decision-making systems is difficult because adversaries are developing counter-AI techniques that exploit inherent weaknesses in AI logic (Ertan & Gray, 2021, p. 219).

1.4 Ethical and Legal Considerations of AI in Decision-Making and Future Challenges

State actors are duty-bound by ethics in war, and the development of AI and its use in autonomous systems – which may be operated at the lowest tactical level – presents a challenge to leadership in the military at all levels (Yan, 2020). The authoritarian and centralised nature of certain governments allows those states greater ability to collect data, domestically and internationally, compared with democratic nations — this is something the DF has to be mindful of as it seeks to understand and utilise AI (Hunter, et al., 2024). Some scholars are concerned that AI “may not account for ethical, humanitarian, or other hard-to-quantify costs” (Davis, 2022).

The Code of Conduct on Artificial Intelligence in Military Systems by the Centre for Humanitarian Dialogue states that there are serious ethical considerations to keep in mind, namely, “there should be justifiable reasons behind the design, deployment, and launch of AI-enabled military systems and (the) development of related AI equipment and technology” (Dialogue, 2021, p. 4). The vulnerabilities of military systems are a critical concern and Popa identifies various attack vectors, including evasion and poisoning attacks, which adversaries can use to manipulate AI decision-making (2022, p. 391). These risks underscore the need for robust cybersecurity measures in military AI systems, particularly as adversarial capabilities evolve and embrace EDTs.

Some of the literature attempts to outline how AI works and the evolution of the different approaches to developing it (Bostrom, et al., 2022, p. 29). Much of the research is content that users of AI should be oblivious to understanding its inner algorithms, citing humanity's faith in deities as a parallel. It is important to distinguish between accountability and transparency. Although transparency concerns one's understanding of the nature and mechanics of AI decision-making, the argument of accountability is very important, as it fosters trust and promotes responsibility for actions.

In their article "Cyber Personhood", Kushwaha *et al.* (2021, p. 276), borrowing from emerging environmental concerns, argue the concept that certain vital crucial cyber systems — those essential to everyday living (e.g., GNSS) — should have their status protected and thus afford "rights" under national law.¹⁵ The argument is convincing realising that so much of today's modern communication infrastructure is dependent on the network time protocol. Could it soon be the case that AI systems would be considered for such "rights" – analogous to how trees have a protected status under law (p. 279)?

Greipl (2024) advocates for improving technical literacy by the military to critically engage with the outputs from AI. Training programs for military planners and soldiers would be required, and they should focus on orientating users with AI interfaces and increasing their understanding of and how to critically analyse and question automated recommendations. Such initiatives will help ensure that human oversight will remain effective. There are limitations to the level to which a learning AI system can reach. Dreyfus and Dreyfus (2005) stress the importance of the emotional experience that humans must endure to achieve an 'expert' level of knowledge about a particular subject. It is only after feeling anguish and persisting through setbacks and failures that humanity can advance to higher levels of knowledge. It is an emotional experience which an AI system can never replicate.

The use of AI in the planning and conduct of military operations must comply with conventions on IHL. Issues surrounding proportionality, distinction and accountability must be carefully evaluated to ensure that AI-driven decisions align with legal and ethical standards as they do with COs (O'Grady, 2022). Currently, there is no universally accepted regulatory framework governing military AI, creating legal and ethical uncertainties in its deployment (Ertan, 2022, p. 219) though a clearer picture is emerging due to ongoing study (Ertan & Gray, 2021). However, as in cyberspace, it may be the case that different global powers will adopt

¹⁵ Global Navigation Satellite System

varying, indeed opposing, views on AI. NATO and its partners recognise the application of International Law in cyberspace (DFA, 2023) whereas others have opted for a more nationalistic view (e.g., the Great Firewall of China) (Raud, 2016, p. 6). Algorithmic bias remains a core issue in AI governance, as training data will reinforce systemic discrimination in target selection and operational decisions (Mayer, 2023, p. 27). The primary challenge, however, is to balance innovation with accepting responsibility, therefore ensuring that AI continues to enhance rather than undermine human agency in the military decision-making process.

1.5 Conclusion

The deployment of AI and autonomous systems in the DF presents considerable ethical and moral challenges for military leadership today. The swift advancement of AI technologies, although offering improved operational efficiency and strategic insight, prompts significant concerns around the transfer of decision-making responsibilities to algorithms. This chapter examined how military leadership must confront the ethical challenges of C2, targeting and the potential collateral effects of autonomous systems. These issues require a transformation in leadership strategies, including a profound comprehension of both the technological proficiencies of AI and the ethical obligations associated with military operations. Commanders must create a culture of accountability, transparency, and ongoing ethical assessments to ensure that AI is developed and used in alignment with the DF's organisational values. As AI advances, leadership frameworks must evolve to maintain human oversight in military operations and uphold ethical standards. The DF must ultimately combine technical innovations with ethical responsibility, ensuring that AI functions as a tool to augment – not supplant – human judgement in military missions and operations.

It is inevitable that the DF will engage with AI in the future. The human interaction between AI and military C2 systems will be of utmost importance and will impact leadership. Is there a risk that AI will undermine human authority and the responsibility of leadership? This chapter proposes that human oversight remains critical in AI-driven military decisions and that compliance with IHL is of the utmost importance regarding AWS.

CHAPTER TWO: AI AND MILITARY TARGETING

“Who Dies?”

2.1 Introduction

The role and principles of IHL are key considerations guiding the discourse about the use of AI and AWS. This chapter will examine how AI systems can be used in targeting, and we will also examine the importance of IHL. AI technology consists of real-time technical decision-making and has the potential to revolutionise C2 in the military. One of the key applications of AI is intelligence, surveillance and reconnaissance (ISR). AI algorithms can process vast amounts of imagery and sensor data, picking up potential threats or targets far more quickly and accurately than human analysts can. In the realm of combat, AI systems can assist with target identification and reduce the chances of misidentification and collateral damage.

AI can be used in military logistics, surveillance, and decision-making. Although AI in military affairs may be considered revolutionary, there are concerns regarding the lack of ethical guidelines. The shift to a digital transformation in the DF presents challenges, particularly regarding the ethical use of AI (DOD, 2023). There are concerns that AI can violate civil rights and privacy. To protect the rights and privacy of civilians and soldiers, clear rules must be established, particularly in multinational missions (Cymutta, et al., 2022). Biased algorithms may be an issue. AI algorithms learn from data and make decisions, and the data upon which AI systems are trained could potentially skew these algorithms and their results. Consequently, AI could make biased decisions with severe repercussions and discriminate against certain groups. This is particularly important for the DF, as its overseas missions are UN-mandated and in multinational forces.

Some military applications utilising AI raise ethical questions about the use of lethal force. AI decision-making must comply with IHL, which covers the use of force. In this chapter, the conflicts in Ukraine and Gaza will highlight some of these issues. Transparency matters too. Military operations may often be classified; however, in contrast with this, ethical AI requires transparency. As noted in Chapter One, the EU AI Act (2024) proposes regulations on the deployment of AI for its use in critical systems. Clear regulations around the use of AI in the military are a step towards this goal. These guidelines should ensure civilian privacy and rights (Cymutta, et al., 2022).

As warfare evolves, strategic considerations must adapt to the rapid development of EDTs, reshaping both military doctrine and operational capabilities. Future conflicts will be influenced by innovative technologies such as AI, robotics, and biotechnology, which will redefine battlefield dynamics and C2 structures. The European Parliamentary Research Service outlines how EDTs such as AI and advanced robotics will transform both offensive and defensive capabilities of states, impacting their ability to project power effectively (Bellasio, et al., 2021, p. 24). As technological change accelerates, aligning strategies, resources and tactics will be crucial to maintaining a competitive military advantage.

2.2 AI and Legal Considerations

There are several ethical concerns with AI in the military: what will AI be used for? Will the technology be used for tracking? Who has access to the algorithm? Who can afford to provide the AI technology? AI does not ‘think’ the same way humans do or share our values (Bennett, 2023, p. 246). It is possible that AI will work in a way that is unexpected — of particular concern are biases which may be (unintentionally) inherent in the technology. Venables & Roberts (2021, p. 43) raise the emotive subject of the use of AI and kinetic targeting. At the very centre of the targeting process, the IHL framework puts human control over the use of lethal force (p. 43). IHL is founded on four fundamental principles that underpin the targeting process: necessity, humanity, distinction and proportionality (p. 45). The growing automation of military targeting, however, begs questions about whether handing human control to AI-enabled capabilities could compromise and progressively destroy the IHL framework, hence leaving the modern-day engagement space legally ungoverned and civilians vulnerable (p. 43).

Some states and commentators suggest that AI might enhance IHL compliance in armed conflicts (Venables & Roberts, 2021, p. 44). The International Committee of the Red Cross describes IHL as “a set of rules which seek, for humanitarian reasons, to limit the effects of armed conflict” (p. 45). The principle of distinction mandates a clear separation between civilians and combatants, as well as civilian and military objects. Only combatants and military objects may be targeted, while civilians and their property are protected under IHL. The principle of proportionality ensures that collateral damage to civilians is not excessive compared to the anticipated military advantage. Military judgement is required to balance these factors before authorising an attack (p. 45).

Those who plan or decide on an attack are responsible for applying the IHL principles and precautionary measures (Venables & Roberts, 2021, p. 46). Feasible precautions must be taken to avoid collateral damage, dictating how and when an attack is conducted. If the military commander assesses that the collateral damage is excessive, the attack should not proceed (p. 46). Issues of distinction and proportionality are subjective and are questions of common sense and good faith for military commanders. It is this absence of human judgement and experience that makes the concept of autonomous AI capability so difficult to reconcile with the IHL framework (p. 46). The challenge of distinguishing combatants from civilians has become increasingly complex, as adversaries often blend into the civilian population and frequently shift between targetable and non-targetable status (p. 49). The lack of a clear accountability framework for IHL violations by autonomous AI capabilities is a significant impediment to the military's use of this technology. Human accountability is a cornerstone of the IHL framework for targeting in armed conflicts, and any dilution of this principle will undermine that framework (p. 51). The key point is accountability in the event of the unlawful taking of life, and this is perhaps where ethical arguments should focus (p. 55).

The law of neutrality defines the rights and duties of belligerent and neutral states during international armed conflicts. Its main goals are to prevent the spread of hostilities and minimise the conflict's impact on non-belligerent states and populations. A neutral state remains uninvolved in the conflict, though it may have an interest in the outcome (Sullivan, 2023, p. 15). Recent developments have challenged the strict applicability of the traditional law of neutrality to international armed conflicts. Besides extensive data sharing, arms deliveries to belligerent states are on the daily agenda of states that do not consider themselves parties to the conflict (Schmuki, 2023, p. 30). In IHL, rules on 'reconnaissance' and 'espionage' address data gathering but do not clarify if or how such data can be shared with belligerents in international conflicts. The legality of obtaining the data does not affect the neutrality of its sharing; a state may violate neutrality obligations even with legally obtained data (Schmuki, 2023, p. 30). State practices largely remain obscure and non-consistent (p. 38). Thøgersen raises the issue of possible international legal obligations of non-participating states hosting individuals conducting malicious cyber operations against Russia orchestrated by the IT Army of Ukraine (2023, p. 39).

The legal review of autonomous AI systems in kinetic targeting is crucial for its alignment with the IHL framework. This review must verify that the technology is not inherently indiscriminate and that it can follow targeting rules applicable to its function

(Venables & Roberts, 2021, p. 47). Given that modern conflict is inherently fast-paced, physically demanding, mentally draining, and stressful, human beings are likely to experience emotions such as fear, exhaustion, and anger that may adversely influence their decision-making (Venables & Roberts, 2021, pp. 54-55). As autonomous AI capabilities will be immune to emotion and respond to events objectively in accordance with their programming, they could actually provide what IHL is seeking to achieve; that is, the best possible protection for civilians and combatants (p. 55).

2.3 AI and Targeting

Targeting in armed conflict is fast-moving and pressured, with short decision-making cycles and an imperfect intelligence picture. Consequently, targeting decisions are judged by the standard of a reasonable military commander and are made based on known circumstances at the time and the information available. If AI-enabled ISR capability continues to develop as anticipated, it will result in the faster production of a more accurate intelligence picture. The reported advances in image and facial recognition and expression analysis will provide greater certainty in distinction, positive target identification, and more precise collateral damage estimates (Venables & Roberts, 2021, p. 53). The steps in the cybersecurity kill chain include reconnaissance, weaponisation, delivery, exploitation, installation, command and control, and the final step is actions on objectives. Surveillance involves continuous monitoring of a target. Reconnaissance gathers information to address specific military questions. Intelligence, the product of these activities combined with other data, aids in military decision-making, including targeting (Venables & Roberts, 2021, p. 48). AI will enable large amounts of information from multiple data sources to be processed and synthesised more quickly and effectively (p. 48).

An objective of AI is to provide users with actionable recommendations that meet the goals of decision-makers (Linkov, et al., 2023, p. 183). Recommendations generated from AI-based approaches hold advantages over human decision-makers through their ability to analyse vast information quickly in an objective and logical fashion (p. 183). Facial expression analysis could help identify hostile intent, such as in the case of suicide bombers (Venables & Roberts, 2021, p. 48). However, humans are biased, unpredictable, and unreliable and make seemingly ‘illogical and impenetrable’ decisions. Military commanders are only held to a reasonable standard, so why would we expect more from machines (p. 55)? Trust in social situations grows based on performance over time, and trust in AI can be developed in the same way (Linkov, et al., 2023, p. 182).

It is important to note that AI recommendations may not account for decision-makers' values or specific mission needs. (Linkov, et al., 2023, p. 183). For example, following a cyberattack, an AI-generated decision engine may recommend disabling an application on the compromised computer system. This action may neutralise the threat posed by the compromised system but could simultaneously endanger a mission, negatively impacting a user's ability to perform critical tasks, or enable the adversary to extend the effect, duration or scope of the cyberattack (p. 183). Explainability refers to the extent that a system's internal mechanics can be explained in terms that are salient to human cognition (p. 184). There are several limitations of explainable AI; while there may be situations in which AI can be explained, some processes are too complex for human cognition (p. 185). A key advantage of AI may well be its ability to avoid human-like behaviour, particularly when that behaviour is not optimal (p. 185). Human understanding is predicated on AI conforming to known cognition processes. AI may arrive at decisions by avenues that are unfamiliar compared to those upon which humans normally rely. Truly benefiting from AI may entail excusing it from the onus to explain itself to humans because such a demand constrains AI to the same values and limitations that have always underpinned human decisions (p. 185). The foundations of decision-maker actions are grounded upon evidence-based data as well as strategic and tactical considerations, which include factors such as decision context, mission needs, and available resources (p. 189). AI systems should provide a level of confidence and sufficient information so that the decision-maker can trust a recommendation (p. 189).

Just as the military seeks air superiority, they now must aim for data supremacy to achieve decisive conditions in modern conflicts. Military commanders must know how to utilise data and deploy it in the delivery of operational effects — they need to know how to use data as a military asset in the engagement space. To improve the confidence that users have in automated tools, either ML models need to achieve near-perfect results or the general approach to software development needs to be adapted (Strickson, et al., 2023, p. 284). Human-AI tension can be avoided by creating a more effective, ethical, and transparent process that combines the decision-making needs and abilities of both actors (Linkov, et al., 2023, p. 194). Popular culture often frames AI and cyber capabilities as “silver bullets”, which are both effective and multi-purpose. Offensive cyber capabilities are often seen as “force multipliers” with high precision, global reach, relatively low cost and potentially high impact (Smeets, 2018, p. 98). COs, and by extension the use of AI, in conflict are often, but not exclusively, targeted against opponents' military infrastructures such as headquarters, C2 and weapon

systems (Schulze, 2020, p. 184). Such operations stand in contrast to the use of strategic cyberattacks in peacetime to influence the decision calculus of adversaries (Smeets, 2018). Strategic cyberattacks typically target sources of national power or society in general (Libicki, 2009, p. 117) but can also be used for tacit bargaining, coercion or deterrence (Schulze, 2020, p. 185). With the advent of 5G communication technology and a complete redesign and rethinking of mobile telephony networks, the attack surface will increase dramatically. This will affect military C2 and mobility (Pernik, et al., 2021, p. 6). Software supply chain attacks can also target telecommunications technology, including 5G networks - for example, software defined networking introduces vulnerabilities to the core network, which may be exploited by malicious actors (p. 18).

Network security threats target the following dimensions of 5G architecture: wireless communication, end-user devices, and 5G infrastructure components (servers, base stations, cells, etc.). Threats to wireless communication fall into four categories: eavesdropping and traffic analysis, jamming, DDoS and adversary-in-the-middle attacks.¹⁶ The major risk to, and threats against, 5G infrastructure are virtualisation, network slice security, improper access control, data replication, roaming partner vulnerabilities, and DDoS attacks (Pernik, et al., 2021, p. 22). Threats against end-user devices — such as targeting user data and privacy (e.g., IMSI catching), and signalling-based attacks — should also be considered (p. 23).¹⁷ The goal on the operational level is to obtain advantages over the enemy in a series of battles (Schulze, 2020, p. 185). Most COs are intentionally designed to stay below the threshold of an armed attack to avoid escalation into conventional conflict (p. 186). One benefit of strategic use of COs and AI is having another option on the table in the game of multipolar geopolitics (Smeets, 2018, p. 97).

An issue of cyber in conflict is that strategic cyberspace capabilities are target-dependent and need to be tailored to specific target configurations. Because malware must be custom-built, it is more difficult to have up-to-date stockpiles once conflict occurs. Missiles and munitions are built once and are ready to use during their expected shelf life of 30 years; however, zero-day vulnerabilities have a shorter life cycle and cannot be stored in the same way (Schulze, 2020, p. 187). High-value targets, such as critical infrastructures and C2 systems, are often air-gapped and require specialised intelligence to gain access. This approach requires time-consuming social engineering to gain a foothold on a system and implies high operational

¹⁶ Distributed Denial of Service

¹⁷ International Mobile Subscriber Identity

complexity for a vast-scale strategic operation (p. 188). Operation Orchard took place in Syria in 2007, in which IDF hackers disabled a Syrian anti-aircraft radar in Tell Abyad and then launched a kinetic air strike. The Israeli Air Force then destroyed a nuclear test site in northern Syria. The operations were successful, and the cyberspace component played a significant role in allowing Israeli F-15 jets to enter airspace unnoticed. This is an example of the sequential use of COs as an “enabler” for kinetic operations — the CO producing an effect that is necessary for a subsequent kinetic operation (p. 189). Large-scale COs with a cyber-physical component, such as Stuxnet, must be tested in simulated environments.¹⁸ However, if there is no time for testing and it is difficult to assess the impact of a COs, military planners are hesitant to use them (p. 190). As in the case with NotPetya, there is a risk of losing assets because of malware spreading in an uncontrolled fashion (p. 191).¹⁹

Information Operations (IO) may be used to facilitate the kinetic targeting process. Though Venables & Roberts (2021, p. 52) acknowledge that ML technology can display preferences, they still commend the benefits, with respect to IOs, of AI’s ability to analyse large amounts of open-source online information to understand how to influence target audiences and tailor messaging to them (p. 48). The process may involve ensuring that a target is in a desired location at a particular time or that an area is clear of civilians (pp. 48-49). In respect of IOs, militaries ought to be aware of the amount of content that is being harvested by social media platforms and where it is being stored and by whom. For example, TikTok is a classic case of combining an obscure background and data collection for the purpose of developing new capabilities (Estonian Foreign Intelligence Service, 2024, p. 76). The massive collection of data provides ByteDance, the owners of TikTok, with the opportunity to develop AI (p. 77). In addition, the data collected may also be useful for other purposes. TikTok extensively gathers information about the device and its user, including contacts, calendars, other applications, Wi-Fi connections and location. Such information can be valuable for intelligence gathering, extortion and cyberattacks, as it can be used to craft convincing phishing emails tailored to a specific individual or their employer (p. 77) and had been highlighted in an earlier paper by Kaska (2019, p. 11).

¹⁸ Stuxnet: used in 2010 to sabotage the centrifuges in Iran’s nuclear facilities, it is widely regarded as the first known “cyber weapon”,

¹⁹ NotPetya: destructive malware attacked targeting Ukraine in 2017 and caused global disruption to businesses.

2.4 AI and Current Conflicts

Two major military confrontations currently taking place are the war in Ukraine and the conflict in Gaza. An examination of the use of AI in these conflicts and what lessons can be identified and learnt will be of vital importance to the DF. In Ukraine, there has been significant use of drones, and Palantir has played a defining role in the Ukrainian defence and planning of military operations (Bergengruen, 2024).²⁰ The IDF has used AI for targeting against Hezbollah in Lebanon (Chmouny, 2025) but also in Gaza (Sylvia, 2024). This situation highlights the absence of, and the necessity for, regulatory frameworks to govern the use of AI in military conflicts. The issue is quite pressing — there are thousands of individuals located in non-participating states who have participated in hostile cyber activities against Russia, orchestrated by the IT Army of Ukraine (Thøgersen, 2023, p. 53). Those states appear to prefer to refrain from taking action to prevent individuals on their territory from engaging in such operations, triggering the question of the possible obligations of the territorial state (p. 53). This is the dilemma: the application of IHL to COs conducted by the IT Army of Ukraine involves complex legal issues regarding the responsibilities of non-participating states whose territories are used in these operations. Greater clarity requires more states to express their views, as nations play a key role in defining the cyber-specific aspects of international legal norms (p. 54).

A distinct feature of the aggression experienced by Ukraine has been cyberattacks on their critical national infrastructure (CNI) — this can only be exacerbated with the advent of AI-enabled tools. The use of Supervisory Control and Data Acquisition (SCADA) systems is typical for most CNI. There has been an increase in the number of queries about SCADA systems in Ukraine. Such an escalation indicates the activity of hackers in the reconnaissance stage of a cyberattack (Davydiuk & Zubok, 2023, p. 129). Cyberattacks on energy companies are increasingly complex and difficult to detect. Vendors providing hardware and software to these companies face constant threats, making supply chain attacks a growing concern (p. 123). Ukraine has shown resilience and is adapting, procuring, and widely installing alternative power sources, backing up data in cloud storage both in Ukraine and abroad, and training personnel (p. 127). Many expected Russia to launch waves of sophisticated cyberattacks beyond Ukraine, specifically targeting the USA and other NATO nations to retaliate against economic sanctions (p. 87). While the Ukraine conflict has witnessed a significant volume and

²⁰ Palantir: software used for integrating, managing and analysing large datasets for governments and militaries.

diversity of cyber incidents, research indicates that the cyber dimension of the war did not play out as analysts initially expected (Loneragan, et al., 2023, p. 85). However, a cyberattack that is consistent with that assumption is the Viasat telecommunications hack, which took place on 24 February 2022, just one hour before Russian forces crossed into Ukraine. The evidence suggests that the Viasat cyberattack was part of Russia's plan for the initial phase of the campaign designed to paralyse Ukrainian C2, likely as part of a decapitation strategy, through both kinetic and non-kinetic means (p. 90).

Evidence of Russian cyberattacks against Ukrainian CNI, including power generation plants, suggests attempts at a more significant cyber campaign. However, these fell far short of the magnitude anticipated prior to the conflict. For example, a cyber incident took place on 8 April 2022, when Ukraine's power grid was targeted by Sandworm, the notorious Unit 74455 of Russia's Main Intelligence Directorate (GRU) (Davydiuk & Zubok, 2023). The attackers had gained access to the power grid's system months earlier, highlighting the fact that COs take time to plan and execute. Had this malware been successful, it would have deprived roughly two million Ukrainians of power. Ukraine's national Computer Emergency Response Team, with international assistance, was able to thwart the attack in progress. Interestingly, the malware used in this incident was comparative to that deployed in 2015, which succeeded in briefly knocking out power for 100,000 Ukrainian citizens (Loneragan, et al., 2023, p. 91). Could it be the case in the future that a polymorphic, AI-enabled malware would be more effective?

Most cyber activity in Ukraine consists of intelligence operations or disruptive attacks, which have limited and temporary effects, with targets quickly recovering and restoring functionality (Loneragan, et al., 2023, p. 98). Nonetheless, policymakers should anticipate that future conflicts are likely to have a third-party cyber component. The fact that most of the cyber spillover in Ukraine has been conducted by these actors, rather than directly by a government, raises questions about how to prepare for and defend against this kind of activity and ensure guardrails are in place to prevent these attacks from inadvertently escalating conflicts (p. 99). Of particular note is the location of these third-party cyber contributors; once the existence of the armed conflict is established, nearly all direct state COs in support of Ukraine would violate the traditional neutral state obligations of non-participation and impartiality (Sullivan, 2023, p. 20). Western corporations who have provided Ukraine with cybersecurity services, software, and equipment worth tens of millions of dollars similarly conform with the law of neutrality despite their potentially important role in defending Ukraine's military and civilian networks

and infrastructure (p. 19). Does this have implications for the Microsoft volunteers who assisted Ukraine during the conflict?

COs, with the exception of DDOS attacks, typically require months of planning, are ambiguous and their effects almost always temporary, and they are challenging to integrate into military planning — many nations see them as strategic assets (Libicki, 2020, pp. 205, 211). AI programming will allow malware to alter its tactics in line with mission parameters as it learns about the environment in which it is operating, along with the defenders and users that populate it (Whyte, 2020, p. 223). Lessons identified in Afghanistan and Iraq show that in difficult environments, such as vast landscapes and deserts, technology tends to fail. For COs and AI tools to work at the tactical level, a data connection with enough bandwidth must exist. Communication equipment needs power, and when that is in short supply, they tend to be unreliable in combat situations. For AI operations in the field, a certain proximity to the target is usually required – an enemy WLAN can only be hacked within radio wave range.²¹ Therefore, tactical COs in the field only make sense if there is spatial proximity, if the required expertise is not too high, and if the effects can be limited to the local proximity (Schulze, 2020, p. 192).

Since the effect of cyberattacks is often temporary, there is a risk of unanticipated resilience (Schulze, 2020, p. 194). The broad conclusion is that conflict in cyberspace rarely echoes into the world of kinetic conflict and that incidents of cyberspace conflict are often *sub rosa* and usually unacknowledged (Libicki, 2020, p. 200). Finding malware in a target system may offer little indication of what its purpose was: e.g., espionage or sabotage? (p. 202). Libicki concludes that a kinetic retaliation to a cyberattack is possible but cannot yet be deemed a likely consequence (p. 203). Therefore, while the non-lethality of cyberattacks means that a plausible response would be non-lethal, the failure to respond to NotPetya suggests that the broad scope of a CO also plays a role. Perhaps a cyberattack that damages software and thereby levies costs on victims is different in kind from a comparably costly kinetic attack that damages hardware (p. 204). These are the lessons Cyber Commands must grapple with right now and learn fast. The more data available to malicious actors interested in leveraging the advantages of AI for cyber aggression, the more capable the techniques employed might be (Whyte, 2020, p. 222).

²¹ Wireless Local Area Network

2.5 Conclusion

Gray (1993) asserts that the character of war continues to evolve, and future conflicts are likely to see operations in cyberspace using AI. The ubiquity of digital platforms and communication networks has introduced new vulnerabilities, with COs and AI-based weapons systems becoming critical elements of modern warfare. The decentralised and anonymous nature of cyber warfare makes it difficult to attribute attacks, raising new challenges for deterrence and defence. The Russian DDoS cyberattack on Estonia in 2007 serves as a prime example of how COs can destabilise nations without traditional military engagement, highlighting the shift toward non-conventional forms of conflict. Currently, the destructive use of COs in Ukraine and the egregious use of AI for targeting by the IDF have given DF leadership copious amounts of data for deep consideration and reflection. Developments in cyberspace and AI create new challenges for conflict prevention. The influence of rising powers in a multipolar world also complicates global security. To prevent future wars, a combination of regulatory frameworks underpinned by IHL, deterrent strategies, collective security mechanisms, and economic cooperation is essential.

International institutions such as the UN, EU, and NATO, as well as the DF, must continue to adapt to new forms of conflict, including hybrid warfare and AI-enabled COs, while also addressing the root causes of conflict. Geopolitics and global power dynamics will have a major role in shaping future conflicts, particularly as the world shifts from a unipolar to a multipolar order. As will be examined in Chapter Three, the strategy and policies surrounding collateral effects and attribution will play a significant role in shaping the balance of power in a new multipolar world. Technology will shape the future of warfare, altering the way states compete and safeguard their interests. EDTs, including AWS, space-based technologies, and AI, will impact military conflict from now on, enhancing both surveillance and combat capabilities. These technological developments will require states to adopt new military and diplomatic strategies.

CHAPTER THREE: AI AND COLLATERAL EFFECTS — ACCOUNTABILITY AND ATTRIBUTION

“Who Cares?”

3.1 Introduction

The work of integrating AI and COs into C2 systems and military decision-making processes is challenging. Some of the most profound impacts are not only tactical but also ethical, legal and political. Further complicating the situation is that AI, deeply embedded into autonomous systems, does not necessarily equate to autonomy in and of itself (NATO CCDCOE, 2021, p. 53). If it were, or rather when it becomes, the case, it would make the task of attributing much more straightforward. Quite possibly the idea of cyber-personhood proposed by Kushwaha *et al.* (2021) has significant potential. This chapter focuses on three areas, namely, the collateral effects estimate process (CEEP) and the potential undesired damage caused by AI, the challenges of attribution and responsibility, and the moral impact on military leadership of autonomous systems.

3.2 AI and the Collateral Effects Estimate Process

Due to the characteristics of the physical, logical and cyber-persona layers of cyberspace (see Figure 1), the attribution of malicious cyberspace activity to a specific person, criminal organisation, non-state actor, or even a nation state can be extremely difficult (US DOD, 2018, pp. IV-7). The design of cyberspace and evolution of the Internet facilitate anonymity, and, combined with applications intended to hide the identity of users, attribution of malicious activity will continue to be a challenge for the foreseeable future (pp. I-12). The cyber-persona layer, the aggregate of an entity’s online identities, and an abstraction of logical layer data hold important implications for joint forces in terms of positive target identification (pp. IV-9).

A significant portion of the operational planning and design framework concentrates on achieving decisive conditions by operational effects delivered through operational actions (SHAPE, 2021). Such effects are well articulated and understood by military planners. In respect of targeting and the execution of non-kinetic operational actions, serious consideration must be given to the CEEP as it provides a framework for operational planners and the commanders to ensure that due diligence has been applied to understanding collateral effects in cyberspace (CCDCOE, 2020, p. 61). It is argued that the collateral effects consideration for COs and AI-enabled autonomous systems must be far more comprehensive in nature than the

targeting cycle process (p. 39). The Cyber Commanders' Handbook (2020, p. 61) presents such a CEEP for COs, which may be beneficial for AI-enabled AWS (see Figures 2 – 4). This CEEP has six phases: Target Validation, Propagation, Non-strike Entities (NSE), Persistence, Assessment, and Commander's recommendation.

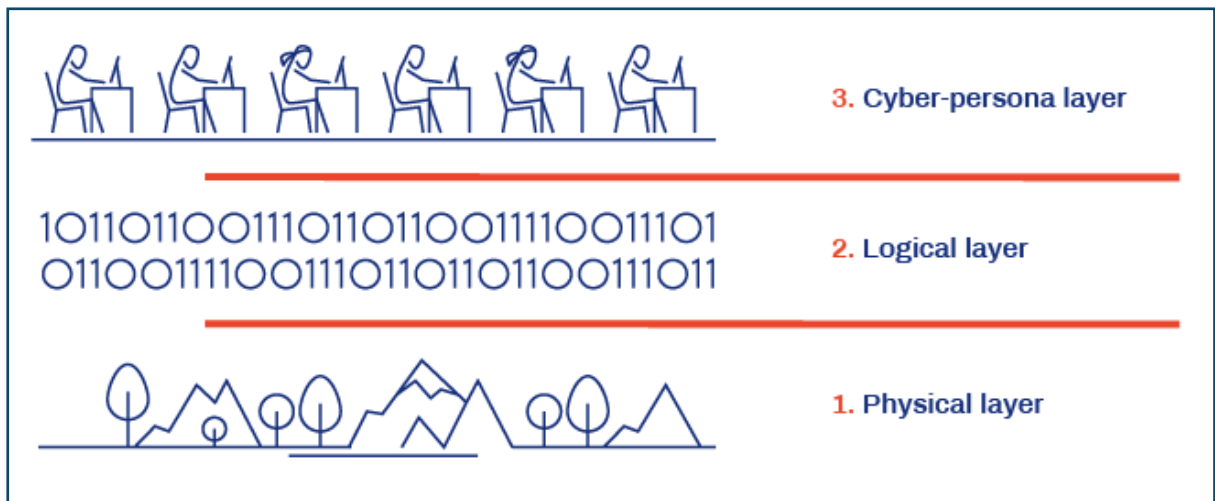


Figure 1: The Three Layers of Cyberspace (CCDCOE, 2020, p. 3)

Phase One ensures the target is valid, with positive identification and Rules of Engagement (ROE) authorisation. If these are not in place, further target validation is needed. Consideration is given to any potential environmental or ecological issues which may arise and highlights the sensitive nature of the target. Consideration is also given to the status of the proposed target — e.g., if it is included on a predetermined NSE list, higher authority approval is required and removal from the NSE list before validation (p. 61). The assessment of the residual risk is currently a staff task. A challenge for planners and commanders is to decide whether this assessment can be conducted by an AI-enabled autonomous system without human intervention or oversight.

Phase Two involves evaluating the propagation of the effect beyond the intended target. If the planned effect cannot extend outside the target, the collateral effects estimate value is zero, marking the completion of the CEEP. Conversely, if the effect can extend beyond the target, an estimate of its propagation extent is necessary. Limited propagation refers to effects that are confined to the target system or network only; local propagation pertains to immediately connected networks, systems, or services, and wide propagation includes the former and subsequent indirectly connected networks. This propagation effect may be desirable, as it is a standard TTP for a CO to “move laterally” across a network, remaining

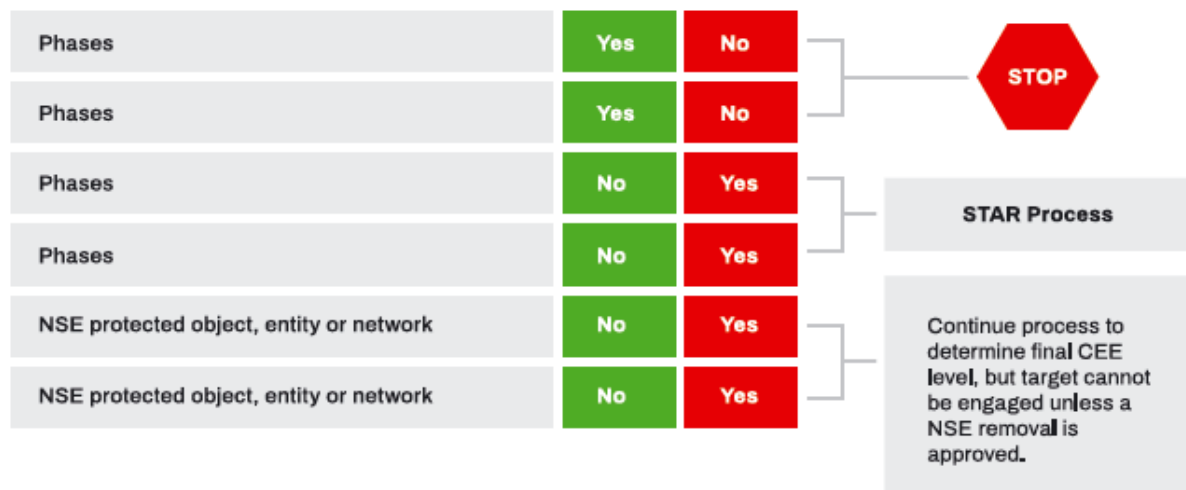
undetected.²² Consequently, the effect may propagate from the target to other legitimate targets, rendering it transient and causing the collateral effects estimate to be zero (CCDCOE, 2020, pp. 61-62).

Phase Three requires consideration of whether the effect could extend to NSEs (p. 62). This is to understand the probability of the effect extending beyond its intended target and having adverse effects. It could be argued that Stuxnet was developed with such considerations, in contrast with NotPetya, which had no such considerations and consequently wreaked havoc on the global economy (Kaminska, et al., 2021, p. 71). The fourth phase considers how persistent the effect is, specifically if the effect can be considered reversible either by the victim or by the perpetrator. Reversible is defined as being possible, through a series of actions, to reverse or mitigate the effect and return all services to normal. In such a case, the effect is both non-destructive and non-permanent (CCDCOE, 2020, p. 62). Phase Five, Assessment, focuses on three critical areas: civilian casualties, the nature of the collateral systems (e.g., single endpoint, networked device, public/private network and/or CNI), and recovery effort (e.g., is the effect reversible?). The final step then is the Commander's recommendation. The CEEP should indicate and highlight the key areas of concern and focus for staff prior to authorising a non-kinetic strike action (p. 64). This process provides a sound template for the military planning staff to successfully integrate AI-enabled autonomous systems into the targeting cycle. However, it does not, at this stage, provide a sufficiently strong template that could be trusted entirely to an AWS. Implementing a CEEP will also assist planners and operational commanders to make a more informed assessment of the performance and effectiveness of an AI-enabled operational actions.

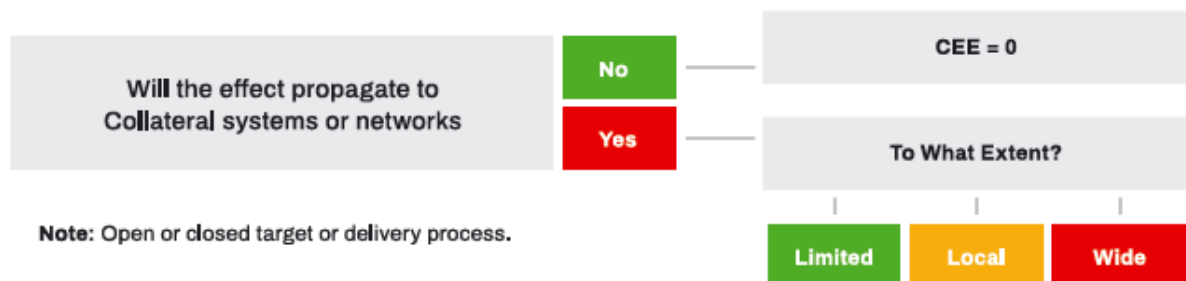
²² Tactics, Techniques and Procedures

Step 1

TARGET VALIDATION



Step 2



Step 3



Step 4



Figure 2: Collateral Effects Estimate Process Steps 1 – 4 (CCDCOE, 2020, p. 58)

Step 5

Estimate the total number of civilian/non-combatant casualties expected as a direct result of the effect and its immediate implications.

Note: Anything above Cdr's Guidance (which might be 0) accrues a 5, no CIVCAS accrues a 1. Within bounds accrues a 3.

CIVCAS

High	5	Above Commanders Guidance
	4	
Medium	3	Within Commanders Guidance
	2	
Low	1	0 Casualties

Estimate the total cost of effort (time, money & manpower) likely to be expended as a direct result of the effect and its immediate implications.

Note: This value will necessarily be an assumption and be formed from an understanding of the collateral system and the effect to be delivered. Persistence once the chosen effect has stopped being delivered on the target is key to this valued judgement.

Estimated Recovery Effort

High	5	Complete replacement
	4	Contractor (High cost & time)
Medium	3	Contractor (Significant cost & time)
	2	Local admin & minimal cost
Low	1	Immediate replacement

Estimate the nature of the systems that are likely to be impacted as a direct result of the effect and its immediate implications.

Note: This value will rely on an understanding of the target system and the collateral concerns surrounding it, both virtual, and physical.

Nature of Collateral Systems

High	10	CNI Sys (functional)
	5	CNI Sys (non-functional)
	4	Large commercial (significant cost & time)
Medium	3	Small / medium enterprises
	2	Non-commercial (small)
Low	1	Personal device

Figure 3: Collateral Effects Estimate Process Step 5 (CCDCOE, 2020, p. 59)

Step 6

COMMANDER'S RECOMMENDATION

Collate the combined score from the collateral effects assessment in order to generate an overall CEE value. Generate a recommendation narrative.

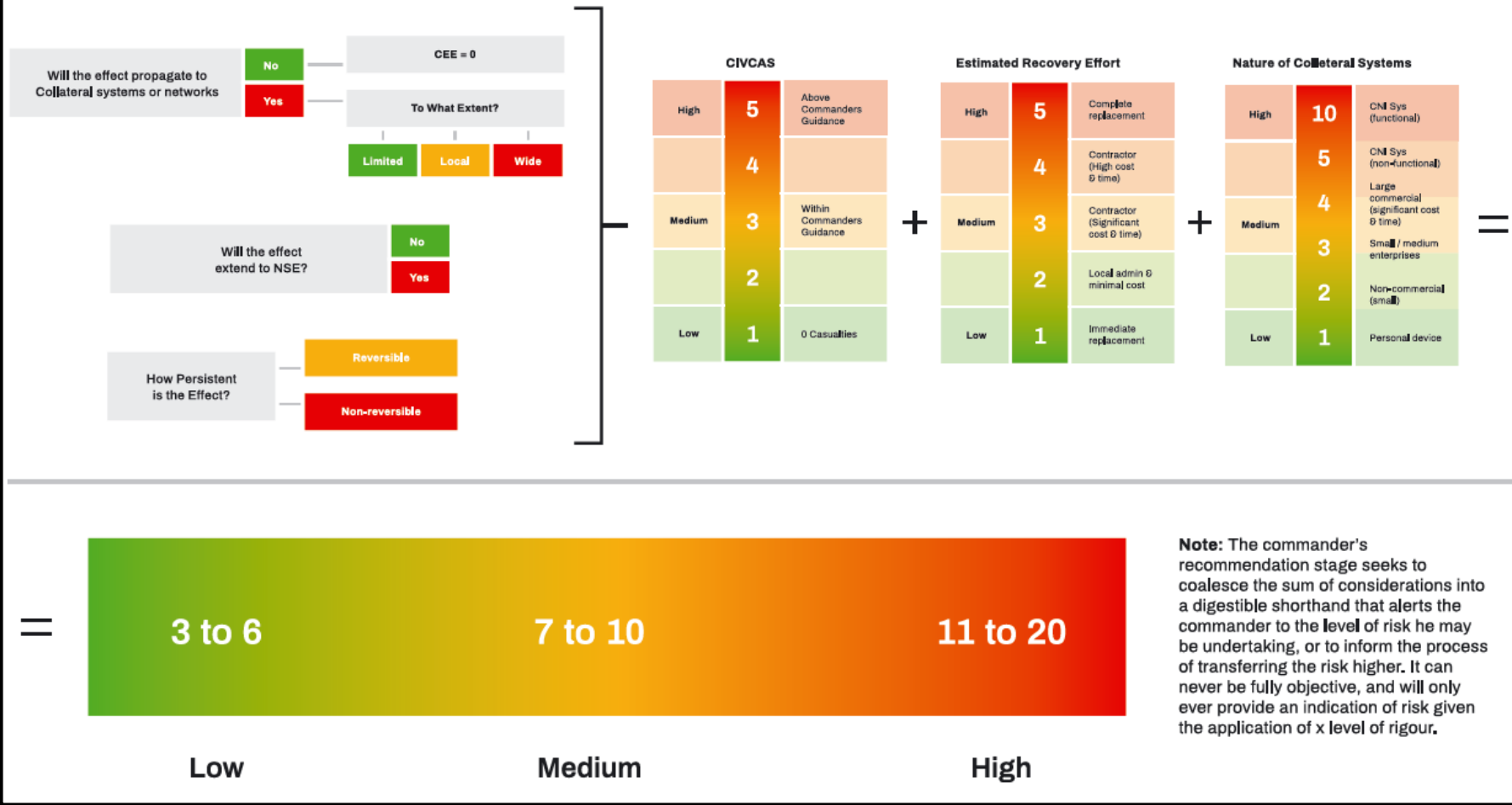


Figure 4: Collateral Effects Estimate Process Step 6 (CCDCOE, 2020, p. 60)

3.3 AI and Attribution

Commanders may never understand how an AI-powered system arrives at a decision, as discussed earlier in Chapter One. Regardless, those leaders are accountable, and this raises concerns about meaningful human control. There is an attribution problem regarding conflict in cyberspace caused by blurred actor identities, plausible deniability, and delayed accountability (CCDCOE, 2020). The problem is that if attribution is unclear or even impossible, retaliation becomes risky or misdirected, creating a dangerous scenario in international crises and conflicts. This all contributes to more uncertainty in an engagement space in which commanders must plan and conduct operations. Senior commanders and strategic leaders must make decisions with imperfect information and limited time, risking miscalculations and mistakes.

Egloff (2020) argues the advantages and challenges of publicly attributing a CO in detail, and it is clear that a public declaration of same is a decision for political leadership (p. 8). Given the apathy of the State towards security and defence, this inertia, in respect of identifying hostile AI-enabled activity on its network, will add another layer of complexity for commanders in the DF (Staunton, 2011). However, it may be the case that the nascent Joint Cyber Defence Command (JCDC) may explore and expose the DF's, and the State's position on public attribution prior to the adoption of AI-enabled systems and AWS, thus providing a tangible framework. Nonetheless, Egloff is clear when they argue that beyond a situational response, deliberate public attribution by policymakers is always a political strategy (2020, p. 7). One of the characteristics of public attribution highlighted by Egloff is that a political decision is being used to shape the engagement space with the aim of establishing and sustaining rules of behaviour (p. 5). Such public statements were normally given in harmony by the Five Eyes community, but given recent shifts in geopolitics and the threat of tariffs to the foreign investment base in Ireland, it seems certain that political leadership will retain control of public attribution of any AI-enabled operation (p. 8).²³

Schmitt *et al.* (2022) offer a very useful “mind map” to assist military operational planners with their military response options to malicious COs, and this will be of assistance as Cyber Commands navigate the peculiarities of responding to AI-enabled COs and autonomous systems (see Figure 5). Irrespective of potential classified advances in technology, it is highly unlikely that cyber attribution will ever rise to anything near the level of attribution

²³ Five Eyes: intelligence sharing community among USA, UK, Canada, Australia and New Zealand

given for, as an example, an ICBM attack (Alexander Klimburg (Ed.), 2012, p. 84).²⁴ That poses the question: how much attribution is enough? Would, for instance, a 50 percent CO attribution be sufficient in a strategic conflict environment? Furthermore, deterrence requires that robust offensive cyber capabilities not only be discreetly available but that the potential adversary also be informed that these capabilities exist (p. 84).

COs encompass a wide range of activities. This means there are different definitions, or legal categories, depending on the character of the CO. These categories are separated by thresholds. In essence, a threshold evolves when a legal framework changes due to a shift in the characterisation of an incident. Thresholds are closely linked to the provisions of the United Nations (UN) Charter and to the problem of attribution. Regarding the UN Charter, only states are subject to its provisions. Therefore, without a positive attribution to a state, the Charter is not applicable. In such cases, assumptions are likely to be that the perpetrator is an individual or non-state actor, and the CO will be labelled as cybercrime or cyber terrorism, which calls for the application of criminal law, not the UN Charter (Alexander Klimburg (Ed.), 2012, p. 168). Establishing attribution for COs is difficult but not impossible. No simple technical process or automated solution for determining responsibility for COs exists. Often, the painstaking work requires weeks or months of analysing intelligence and forensics to assess culpability. In some instances, the international community may establish attribution for a CO within hours of an incident, but the accuracy and confidence of the attribution will vary depending on available data (ODNI, 2018, p. 2). Attributing an attack to a particular country or actor requires collecting as much data as possible to establish connections to online actors, individuals, and entities. The primary three indicators are tradecraft, infrastructure, malware, and intent (p. 3). Attribution assessments typically include a series of judgements that describe whether the event was an isolated incident or not (p. 4). Attribution involves identifying or locating an attacker or intermediary, usually after an attack is detected. This process can occur while the attack is ongoing or after it ends. Defensive COs prioritise mitigating attacks over identifying responsible parties, but attribution is crucial for reciprocal actions and offensive COs (CCDCOE, 2020, p. 22).

²⁴ Inter-Continental Ballistic Missile

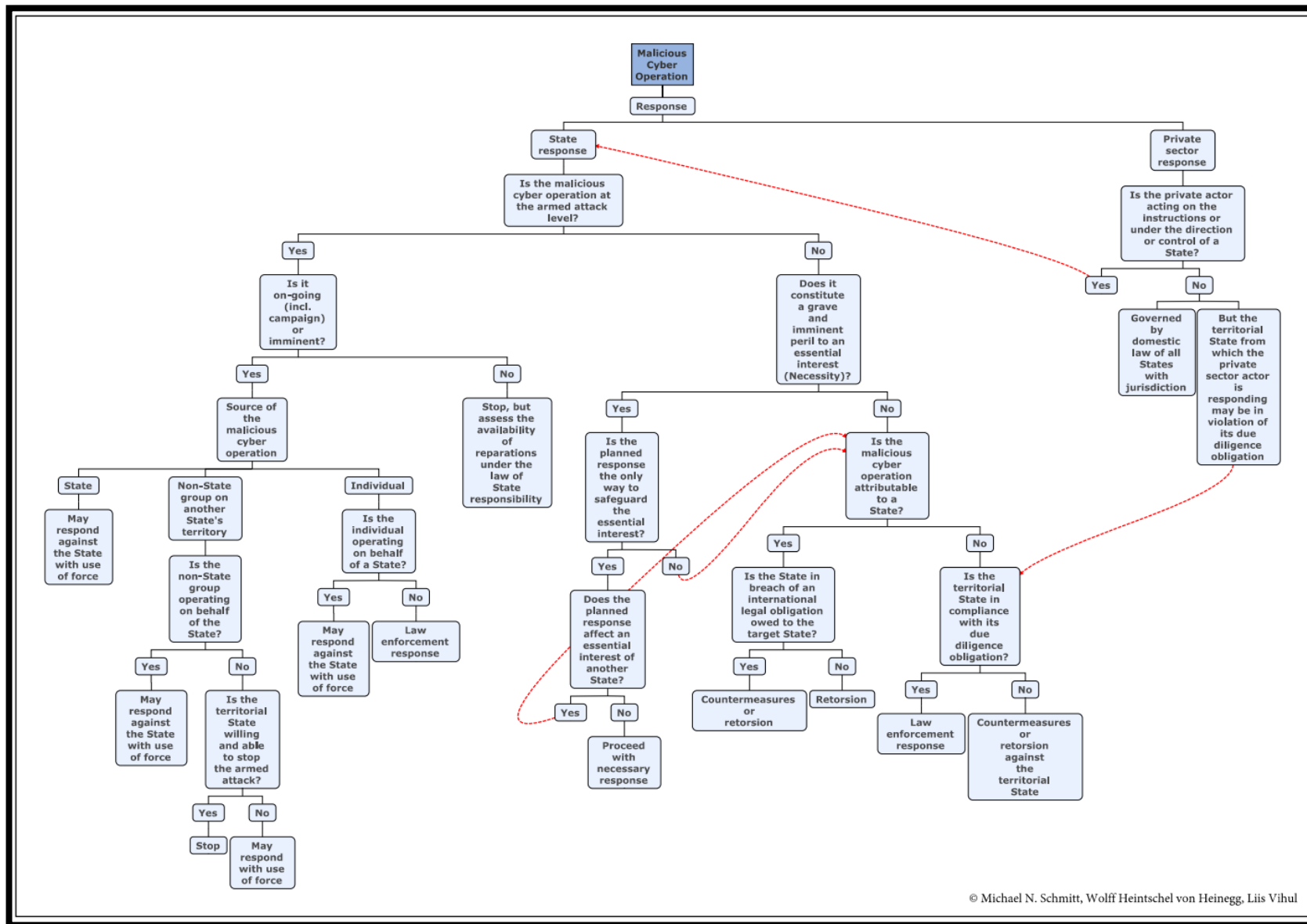


Figure 5: Flow Chart of Malicious Cyber Operations Response Options

Attribution has intelligence, technical, legal, and political aspects, and a military Cyber Command may be involved in each of them. The level of involvement varies by nation, and as stated earlier, public attribution is often a political decision. Threat actors often reuse known TTPs, complicating attribution (CCDCOE, 2020, p. 22). Attribution relies heavily on cyber forensics and threat intelligence, with sources ranging from open-source intelligence (OSINT) to technical analysis and closed sources. Actors in cyberspace seek anonymity, using obfuscation and false flag operations and proxies; therefore, attribution must be supported by evidence from multiple sources (p. 22). Cooperation with public-sector entities such as government CSIRTs and law enforcement, as well as private-sector partners, can enhance capabilities for attribution.²⁵ Acts by state organs are attributable to the state, including actions by non-state actors under state direction or control, or when the state adopts their conduct (p. 22). Since most COs originate in other countries, cooperation at the international level is essential. International cooperation is necessary to repel, identify, attribute and respond to COs against protected networks and systems (p. 47). Punishment requires developing the ability to quickly and accurately attribute cyberattacks (CSC, 2020, p. 3).

Challenges in establishing timely and accurate attribution can weaken cyber deterrence by generating doubt about the identity of the perpetrator of a cyberattack and undermining the credibility of response options (CSC, 2020, p. 27). End-to-end encryption is currently impeding governments' ability to obtain lawful access to electronic evidence in investigations ranging from cyber intrusions and attacks to crimes threatening serious harms, such as child exploitation, gang violence and drug trafficking, and domestic and international terrorism. Attributing responsibility for malicious cyber activity to actors—a necessary precursor to many law enforcement responses—can be difficult and slow without access to relevant encrypted data content (CSC, 2020, p. 95).

Attribution is the essential step in justifying any response to a CO, AI-enabled or otherwise. The success of attribution depends on its purpose, which varies based on the actor performing it. Trends in attribution among NATO and non-NATO countries indicate that a higher density of traditional alliances, intelligence-sharing agreements, and aligned incentives correlates with a greater likelihood of coordinated attribution efforts. Finally, the application of IHL in cyberspace remains under development, and establishing evidentiary standards for attribution should be the next priority (Derian-Toth, et al., 2021, p. 20). Effective attribution

²⁵ Computer Security Incident Response Team

relies on public-private partnerships, collective efforts, evidentiary standards, and clear goals. In comparing private and public sector attribution, robust private sector involvement and strong public-private partnerships result in improved outcomes. Proper attribution can deter future attacks and improve over time (p. 20).

3.4 AI and Accountability

Command structures, legal norms and institutional cultures must adapt to enable AI usage within the military. With the possible integration of AI, there is a severe risk that there will be an erosion of traditional command responsibility. For example, how does the accountability chain work when an AI system acts autonomously? Can responsibility be legally delegated to machines, and if so, should the concept of cyber personhood, as proposed by Kushwaha *et al.* (2021), be examined more carefully and given more credence? Programmers, private contractors, commanders and senior leadership require clear lines of responsibility, but these may be blurred by AI developments. It is possible that emerging Cyber Commands may establish guidelines and principles that can help with the integration of AI into C2 systems. The principles employed by Cyber Commands are based on joint functions from contemporary doctrine, which have limited understanding of multi-domain operations (MDO). Institutions may overtrust machines and become risk averse, shaping how senior leadership roles act and how our moral courage is exercised. The quotation at the start of this thesis states that the sword is not the killer but rather the person who wields it. This is the case with AI, and therefore it would be remiss of the DF to disengage from AI in its military systems. It may be the case that the marketplace decides for, and forces the hand of the DF, as increasingly more and more IT/social media products have some variation of AI embedded in them. As PME has evolved over the years, innovations in military training will continue, and responsibility for design approaches and command policies which reinforce accountability in human AI teams are a necessity (O'Hara, 2016).²⁶

Many countries, including Estonia, France, Germany and Australia, have all affirmed that there is no legal obligation for a state to make public their decisions on attribution. From their respective positions, it is a national prerogative at the discretion of each state (CCDCOE, 2025). As a rule, the conduct of state organs is attributable to the state in question. In contrast, the conduct of non-state actors can only be attributed to the state under specific circumstances (CCDCOE, 2025). This may prove to be problematic, indeed a dilemma for militaries in respect

²⁶ Professional Military Education

of AI-enabled COs. The terrain of cyberspace is dynamic and constantly evolving as new technologies and platforms emerge. Furthermore, COs may be difficult to trace, and despite technological developments, many incidents are likely to be deniable and untraceable (NATO, 2020, p. 13). Although the attribution of activities in or through cyberspace is essential, the identification of actors does not solely depend on digital information. A combination of multi-source intelligence, forensics and other methods all contribute to revealing an actor's identity (p. 13). Non-attributable attacks increase uncertainty and misperception, leading to a perceived reduced political risk and reducing response options. The possibility of COs conducted by a proxy, or under false flags, must also be considered (pp. 13-14). The dilemma here is self-evident; if a nation has been attacked by an AI-enabled CO, and the effects are so severe, the victim nation may wish to raise the possibility of invoking a collective defence article (e.g., EU/NATO). However, if the question of "*Who done it?*" remains unanswered, then it is almost impossible to invoke such collective defence (e.g., Article 5 of the Washington Treaty for NATO). It is both challenging and important for a victim state to be able to attribute an attack suffered. Equally, however, it is very tempting for militaries to develop non-attributable COs, a capability which AI will enhance. Even if such COs do not remain below the threshold of war, if they are non-attributable, the response and countermeasures from the victim nation (kinetic or non-kinetic) will be muted. Is this the type of behaviour that the DF seeks to conduct?

The attribution problem arises because of the anonymity of cyberspace. Attackers cloak their actions by routing them through a chain of intermediaries located in various third states. If investigations are not conducted in an extremely expedited manner, traces may be lost, and it might be impossible to determine from where the source or indeed any of the links in the chain originated. If it is almost impossible to ascertain where attacks originated from, a counterattacking victim might easily find itself barking up the wrong tree, causing tensions with unwitting and innocent proxy parties which could be located anywhere, ranging from hostile to neutral to allied nations, and intentionally engineer a war. Investigations eventually would need to be run through third states. All this makes attribution very difficult and requires extensive forensic and legal cooperation among the countries concerned. Even if the source of the CO can be determined with certainty, possible counterattack options could involve the infrastructure of third-party countries as well. Attribution is a matter of degree as well as political judgement, especially when made public by governments or their agencies. Further

complicating the issue is that disclosing forensic methods and intelligence sources may diminish or even compromise their value for future contingencies (Missiroli, 2021, p. 9).

Public attribution of a malicious CO exposes vulnerabilities, entails reputational damage and elicits some form of retribution, presumably with tangible consequences for the perpetrator. Accordingly, the lack of visible or credible responses may inflict reputational damage on the attributor. Attribution is a form of strategic communication: it is about messaging (bilaterally and discreetly, or jointly and publicly), and it is about perceptions. It requires credibility at the source, including the capability to retaliate (Missiroli, 2021, p. 9). Legal consequences of misattribution in cyberspace are not sufficiently addressed in doctrine, and the topic appears to pose a challenge for the IHL. This outcome is not desirable since mistaken attribution may contribute to destabilising international peace and security (Stejskal & Faix, 2022, p. 217). Anonymity is a key characteristic of cyber conflict.

3.5 Conclusion

Attribution by individual states can take many forms, including criminal indictments, economic sanctions, technical alerts or official statements; collective attributions mostly take place as a series of coordinated statements or press releases by several states. For instance, in December 2017, the UK, US, Australia, Canada, New Zealand and Japan issued coordinated statements attributing the WannaCry ransomware attack to North Korea (White House, 2017). Similar coordinated attributions followed the NotPetya cyberattacks and the Russian cyberattacks against Georgia in 2018 (Roguski, 2020). Public attributions alone, if not followed by enforcement action, do not infringe a state's rights under international law because international law does not prohibit one state from commenting on another state's actions if such comments do not amount to coercion regarding the other state's internal affairs. These public attributions do not constitute internationally wrongful acts which would need to be justified under the doctrine of countermeasures. At most, they might qualify as retorsions, i.e., reactions which do not interfere with the target state's rights under international law (p. 34).

Attribution has limitations due to false flags, misattributions, and the risk of increased tensions or conflict. Despite alternatives such as active defence, attribution remains the best initial response to state-sponsored cyberattacks. The challenges of attribution in the domain of cyberspace are well documented, requiring technical attribution to identify the location and identity of the cyberspace infrastructure from which an operation originates, political attribution to identify the person behind the infrastructure, and legal attribution to identify a

sufficient legal nexus between the persons behind the operation and a state. The complexity of attribution in the cyber context is compounded by various factors, including the ability for COs to be routed through multiple computer networks in different states and the use of “anti-attribution” mechanisms to hide the provenance of COs (Sander, 2019, p. 368).

CONCLUSION

Key Findings and Recommendations

This research paper intended to thoroughly analyse the ethical complexities and leadership challenges posed by the integration of AI and autonomous systems into the C2 systems of the DF. The study has shown that the application of AI in targeting cycles and COs yields substantial operational advantages but also engenders considerable ethical, legal and doctrinal complications. The thesis's key chapters consistently highlighted the dichotomy between technical efficiency and the persistent necessity for ethical military leadership. Chapter One illustrated how AI systems facilitate rapid decision-making in high-stress situations, potentially undermining the supremacy of human judgement, particularly when issues of algorithmic transparency and data reliance arise. Trust and understanding between humans and autonomous systems are insufficiently developed, and the existing DF leadership doctrine lacks clarity on integrating AI into C2 systems. Chapter Two explained the significant ethical complexities associated with AI and targeting, especially in adhering to the standards of IHL, which include distinction, proportionality, humanity and necessity. Case studies, exemplified by the IDF's deployment of "Lavender" and "Gospel", illustrate that diminishing human participation in the targeting cycle results in ethically dubious consequences, thus undermining the legitimacy of the military instrument. Chapter Three expanded on these issues, exploring the domain of cyberspace, revealing that attributing AI-enabled COs or autonomous cyber effects is fundamentally challenging and complex. This chapter introduced CEEP, which presently supports COs and could be adjusted to address the distribution and uncertainty of impacts created by autonomous systems in cyberspace. Furthermore, to address deficiencies in the response mechanism to both COs and AI-enabled attacks, Chapter Three presented a menu of possible response options as proposed by Schmitt *et al.* (2022).

Further Research

The primary conclusion of the thesis is that the integration of AI and autonomous systems into military C2 systems necessitates an evolution in ethical leadership, IHL compliance and clarity on responsibility and accountability. Juxtaposed with the advancements of AI technologies, improved speed, accuracy and situational awareness — especially in ISR and COs — are the risks of moral detachment, diminished human accountability and obscure decision-making. This is particularly concerning with respect to lethal targeting or strategic response options to COs. DF leadership must be aware of, and embrace, these developments.

Conventional paradigms dependent on human agency must now address technology that can initiate or influence operations without clear rationale or complete operator comprehension. The arguments presented demonstrate that leadership transcends mere supervisory or veto functions; it involves active participation with AI systems, their training data, desired outcomes, and the potential for unforeseen consequences. Existing training, educational and technical resources are insufficient to address these developing complexities. Although there are several policy frameworks and guidelines available for AI (NATO, EU and NCSC), the DF lacks a tailored doctrine that delineates the exercise of ethical military command over AI-enabled COs and autonomous systems. This void renders commanders susceptible — both operationally and legally to VUCA situations further complicated by fast-changing circumstances. The utilisation of AI-enabled COs, particularly subthreshold, can complicate the attribution problem which undermines deterrence strategies and obscures legal and ethical boundaries of accountability. Without a suitable framework, the DF may be inadequately prepared to conduct operations compliant with IHL.

Concluding Remarks

This thesis presents some essential recommendations for managing these risks and maximising the operational utility of AI and autonomous systems. The DF should review and update its doctrine to reflect the ethical nuances and operational challenges presented by AI-enabled COs and autonomous systems. This doctrine must be compatible with national guidance and EU regulations and should align with NATO's policy (NATO, 2021). A dedicated office should be created within the JCDC, possibly as a "Special Advisor". This section would be responsible for certifying AI-enabled systems that would be used in C2, targeting, or ISR functions, ensuring that outputs are auditable and consistent with CEEP. This office would ensure that the DF adheres to ethical standards. Further, DF leadership training should be revised to include modules on human-AI collaboration, algorithm biases, and cognition risks. It is essential that commanders possess both technical literacy and ethical confidence when deploying AI-enabled systems. The CEEP should be fully incorporated into the JCDC and include AI-enabled COs, thus allowing commanders to evaluate the extensive, long-term effects of deploying autonomous systems. A further recommendation is a thorough legal and strategic guiding framework must be developed regarding attribution and accountability of AI-enabled COs. This framework should explicitly delineate thresholds for actions, response options, escalation management and international engagement. Ultimately, there needs to be an investment in AI development within the DF. Depending on COTS

technology for defence purposes, while economical, poses the danger of introducing ethical oversights, vulnerabilities, or opaque behaviours into military operations. By establishing a DF framework, commanders can ensure that AI-enabled systems reflect national ideals and organisational values and will comply with operational constraints. Only through these endeavours and measures can the DF ensure that AI-enabled systems will act as a force multiplier for good instead of being a potential source of strategic and ethical liability.

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